

Battery Short-Circuit and Relay/Fuse Coordination Report

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1. Battery Pack Calculation

Pack Voltage	400 V	Pack Resistance	29.816711 mΩ	Maximum External Short-Circuit Current	13415.296 A
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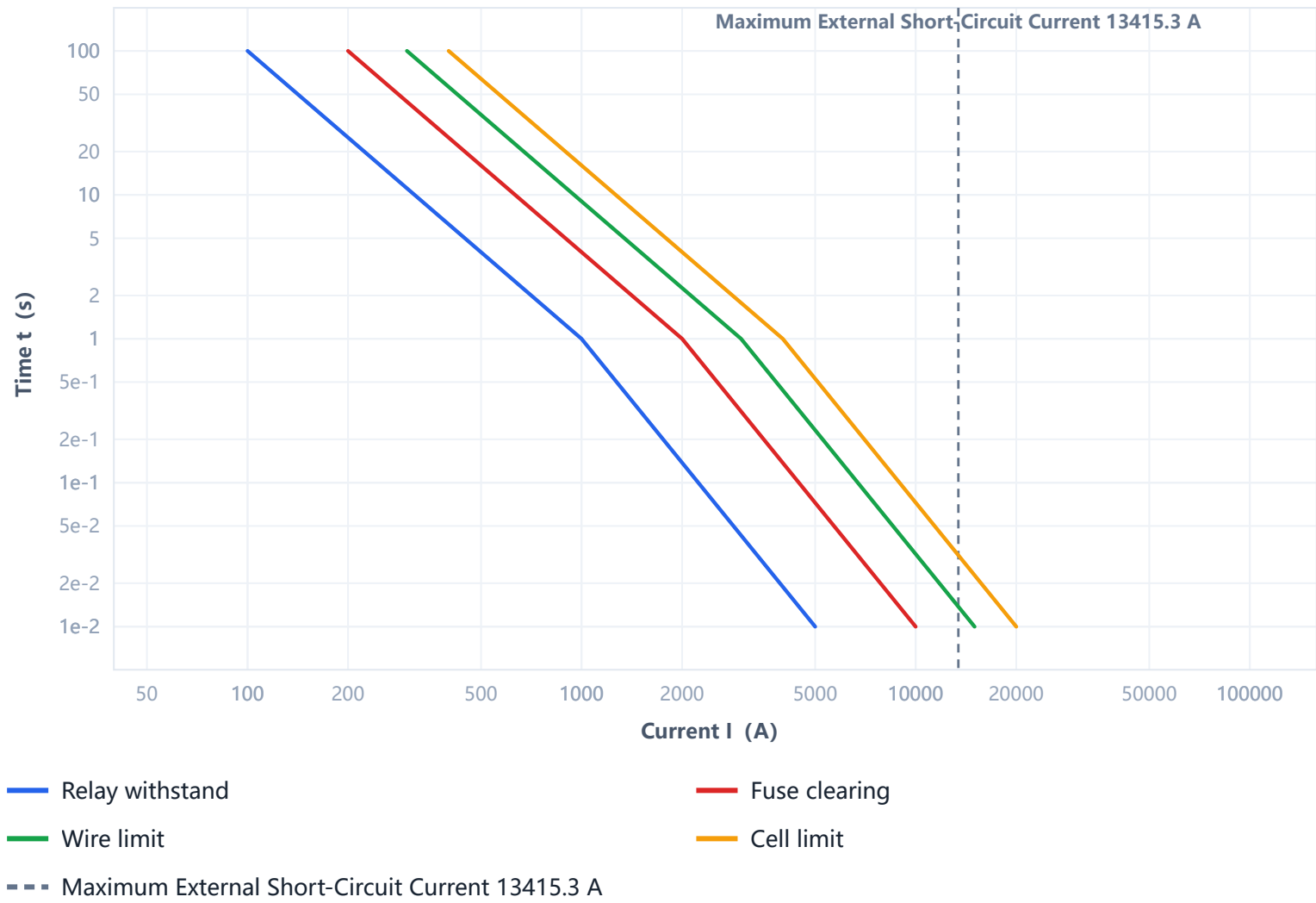
2. Resistance Breakdown

Ref. / Name	Source / Parameters	Single Resistance	Quantity	Connection	Equivalent Resistance
R1 Component 1	Fixed value 0.3 mΩ	0.3 mΩ	96	Series nR	28.8 mΩ
R2 Component 2	Copper Cu (annealed); 80×20×1 mm; 20 °C	0.0688 mΩ	12	Series nR	0.8256 mΩ
R3 Component 3	Copper Cu (annealed); 500×30×3 mm; 20 °C	0.095556 mΩ	2	Series nR	0.191111 mΩ

Network Expression: R1 + R2 + R3

3. Protection Coordination Curves

Protection Coordination (Time-Current)



4. Engineering Analysis

人工校核 <analysis>: retained verbatim

5. Curve Data Summary

Curve Name	Point Count	Current Range	Time Range
Relay withstand	3	100 ~ 5000 A	0.01 ~ 100 s
Fuse clearing	3	200 ~ 10000 A	0.01 ~ 100 s
Wire limit	3	300 ~ 15000 A	0.01 ~ 100 s
Cell limit	3	400 ~ 20000 A	0.01 ~ 100 s